

Unit 4. Radiation

- Introduction to radiation heat transfer
- Kirchhoffs law, Stefan Boltzmann law
- Electromagnetic spectrum, thermal radiation, View factors
- Net radiation between two large parallel planes
- Radiation with shields
- Temperature measurement of gases and radiation errors
- Spectral emissive power, surface emission, total emissive power, emissivity
- Radiative properties, Emission, irradiation, absorptivity, reflectivity and transmissivity
- Concept of black and grey body, radiation intensity
- Laws of black body radiation, non-black surfaces Grey, white and real surface, radiation between black surfaces and gray surfaces

- Heat transfer by radiation usually takes place simultaneously with heat transfer by convection and conduction.
- The heat transfer by radiation is of much more importance at high temperature levels as compared to the other two mechanisms.
- Direct fired kettles, electric heaters, steam boilers, rotary kiln, etc. are the examples of chemical process equipments where radiation is a major energy transfer mechanism.
- **Radiation: It refers to the transport of energy through space by electromagnetic waves.**
- **It depends upon the electromagnetic waves as a means for transfer of energy from a source to a receiver.**
- Radiant energy is of the same nature as the ordinary visible light. It travels in the straight lines and it may be reflected from a surface. T
- The electromagnetic waves with wavelength ranging from 0.50 to 50 μm (microns, 1 μm = 10^{-6} m) are of importance to radiant-heat transfer.
- Radiation of a single wavelength is called as monochromatic.
- Thermal radiation is the energy emitted by a body due to its temperature.
- **In contrast to heat transfer by conduction and convection, radiation heat transfer does not require an intervening medium (material or liquid) between two locations and the heat can be transmitted by radiation mode across an absolute vacuum.**

Typical examples of heat transfer by radiation:

- (i) Transfer of heat from the sun to the earth.
- (ii) Heat loss from an unlagged steam pipe.
- (iii) Use of energy from the sun in solar heaters.
- (iv) Heating of a cold room by a radiant electric heater.

Reference Books:

- Cengel A. Yunnus. "Heat Transfer – A Practical Approach", McGraw Hill.
- Heat and Mass Transfer, K.A. Gavanhe

Absorptivity, Reflectivity and Transmissivity:

- Any substance receives and gives off/emits energy in the form of electromagnetic waves.
- When energy emitted by a heated body falls on a second body (i.e., thermal radiation falling on a body), it will be partly absorbed, partly reflected back and partly transmitted through the body.
- It is only the absorbed energy that appears as a heat in the body.
- The proportions of the incident energy that are absorbed, reflected and transmitted depend mainly on the characteristics of a receiver

[incident radiation = heat absorbed + heat transmitted + heat reflected]

1. The fraction of the incident radiation on a body that is absorbed by the body is known as the absorptivity and may be denoted by the letter 'a'.
2. The fraction of the incident radiation on a body that is reflected by the body is known as the reflectivity and may be denoted by the letter 'r'.
3. The fraction of the incident radiation on a body that is transmitted through the body is known as the transmissivity and may be denoted by the letter 'τ'.

The energy balance about a body (a receiver) on which the total incident energy falling is unity (i.e., the sum of these fractions is unity) is given as :

$$a + r + \tau = 1.0$$

- (i) $r = 0$ represents a nonreflecting surface.
- (ii) $a = 1$ represents a perfectly absorbing surface or a black surface.
- (iii) $\tau = 1$ represents a perfectly transparent surface.
- (iv) $r = 1$ represents a perfect reflector.

- I. If $\tau = 0$, the amount transmitted is very negligible and material is opaque.
- II. If $\tau = 1$, $a = r = 0$, then all the incident energy passes through the body and it is called perfectly transparent, e.g. rock salt (NaCl), quartz and fluorite
- III. If $r = 1$, $a = \tau = 0$, then all the incident energy is supposed to be entirely reflected by the body and is specular.
- IV. If $a = 0$, $r + \tau = 1$, then the body is called as a white body (non absorbing), e.g., a piece of white chalk (white body).

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- Heat and Mass Transfer, K.A. Gavanhe

Black Body:

- A body for which $\alpha = 1$, $\rho = \tau = 0$, i.e., which absorbs all the incident radiant energy, is called a black body.
- It neither reflects nor transmits but absorbs all the radiation incident on it, so it is treated as an ideal radiation receiver.
- It is not necessary that the surface of the body be black in colour.
- The black body radiates maximum possible amount of energy at a given temperature and though perfectly black bodies do not exist in nature, some materials may approach it.
- Lampblack is the nearest to a black body which absorbs 96 % of the visible light.
- The concept of a black body is an **idealisation** with which the radiation characteristics of real bodies are compared. Both the absorptivity and emissivity of perfectly black body are unity.

Laws of Black Body Radiation

Kirchhoff's Law:

- This law sets up a relationship between the emissive power of a body/surface to its absorptivity.
- **Statement of Kirchhoff's law: It states that, at thermal equilibrium the ratio of the total emissive power to its absorptivity is the same for all bodies.**

$$\frac{E_1}{a_1} = \frac{E_2}{a_2} = \frac{E_b}{a_b} = E_b$$

- Consider that two bodies are kept into a furnace held at constant temperature of T K. Assume that, of the two bodies one is a black body and the other is a non-black body, i.e., the body having 'a' value less than one.
- Both the bodies will ultimately attain the temperature of T K and the bodies neither become hotter nor cooler than the furnace. At this condition of thermal equilibrium, each body absorbs and emits thermal radiation at the same rate.
- The rate of absorption and emission for the black body will be different from that of the non black body.
- Let the area of non-black body and black body be A1 and A2 respectively.
- Let 'T' be the rate at which radiation falling on the bodies per unit area and E1 and Eb be the emissive powers (emissive power is the total quantity of radiant energy emitted by a body per unit area per unit time) of the non-black and black body respectively.
- **At thermal equilibrium, the absorption and emission rates are equal.**

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$$I_{a_1} A_1 = A_1 E_1$$

$$I_{a_1} = E_1$$

$$I_{a_b} A_2 = A_2 E_b$$

$$I_{a_b} = E_b$$

$$\frac{E_1}{a_1} = \frac{E_2}{a_2} = \frac{E_b}{a_b} = E_b$$

(As the absorptivity of the black body is 1.0)

- The **emissivity 'e'** of any body is defined as the ratio of the total emissive power E of the body to that of a black body E_b at the same temperature.
- The emissivity depends on the temperature of the body only.
- The emissivity of a body is a measure of how it emits radiant energy in comparison with a black body at the same temperature.

$$e = \frac{E}{E_b}$$

$$e = \frac{E}{E_b} = \frac{a}{a_b}$$

$$a_b = 1 \text{ (for black body)}$$

$$e = a$$

Thus, when any body is in thermal equilibrium with its surroundings, its emissivity and absorptivity are equal.

Monochromatic emissive power: It is the radiant energy emitted from a body per unit area per unit time per unit wavelength about the wavelength λ . It is denoted by E_λ . It has the units of $W/(m^2 \cdot \mu m)$.

Total emissive power: It is the total quantity of radiant energy of all wavelength emitted by a body per unit area per unit time. It is denoted by the symbol E. It has the units of W/m^2 .

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For the entire spectrum of radiation from a surface, it is the sum of all the monochromatic radiations from the surface.

$$E = \int_0^{\infty} E_{\lambda} d\lambda$$

Monochromatic emissivity: It is the ratio of the monochromatic emissive power of a body to that of a black body at the same wavelength and temperature.

$$e_{\lambda} = \frac{E_{\lambda}}{E_{b, \lambda}}$$

Gray body: A body having the same value of the monochromatic emissivity at all wavelengths is called a gray body.

A gray body is the one of which emissivity is independent of wavelength.

Stefan-Boltzmann law: It states that the total energy emitted per unit area per unit time by a black body is directly proportional to the fourth power of its absolute temperature.

$$E_b \propto T^4$$

$$E_b = \sigma \cdot T^4$$

T = Temperature in K

σ = Stefan-Boltzmann constant
 $= 5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$

For a non-black body,

$$\frac{E}{E_b} = e$$

$$E = e \cdot E_b$$

$$E = e \cdot \sigma \cdot T^4$$

- Cengel A. Yunus. "Heat Transfer – A Practical Approach", McGraw Hill.
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where 'e' is the emissivity of the non-black body. The Stefan-Boltzmann equation is a fundamental relation for all the radiant energy transfer calculations.

Heat transfer by radiation:

A body having an emissivity 'e' at temperature T_1 emits the radiant energy equal to $e\sigma T_1^4$ per unit area. If the surroundings are black, none of this radiation will be reflected by them and if surroundings are at temperature T_2 , they will emit the radiation equal to σT_2^4 . If a body is grey, it will absorb fraction 'e' of this energy, so that the net rate of radiant energy flow from the grey body to the black surrounding is given by the expression

$$\frac{Q}{A} = e \cdot \sigma (T_1^4 - T_2^4)$$

'e' = the emissivity of grey body.

T_1 = absolute temperature of grey body

T_2 = absolute temperature of surroundings.

Transfer coefficient for radiation:

The net heat transfer by radiation from a unit surface area of a grey body at temperature T_1 to the black surroundings at temperature T_2 may be given by the relation

$$Q_r = h_r (T_1 - T_2)$$

$$h_r = \frac{Q_r}{(T_1 - T_2)} = \frac{e \cdot \sigma}{(T_1 - T_2)} (T_1^4 - T_2^4)$$

where h_r is the radiation heat transfer coefficient. Equation (4.21) is also applicable if the surroundings are not black, the body is small and none of its radiation is reflected back to it.

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Example: Calculate the heat loss by radiation from an unlagged horizontal steam pipe, 50 mm o.d. at 377 K (104 °C) to air at 283 K (10 °C).

Data : Emissivity, $e = 0.90$.

Solution : The heat loss by radiation per unit area is given by

$$\frac{Q_r}{A} = e \cdot \sigma \cdot (T_1^4 - T_2^4)$$

where

$$e = 0.90$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$$

$$T_1 = 377 \text{ K and } T_2 = 283 \text{ K}$$

$$\begin{aligned} \frac{Q_r}{A} &= 0.90 \times 5.67 \times 10^{-8} \left(377^4 - 283^4 \right) \\ &= 704 \text{ W/m}^2 \end{aligned}$$

Example: A 50 mm i.d. iron pipe at 423 K (15 °C) passes through a room in which the surroundings are at 300 K (27 °C). If the emissivity of the pipe metal is 0.8, what is the net interchange of radiation energy per meter length of pipe ? The outside diameter of pipe is 60 mm.

Length of pipe = 1 m

$$e = 0.8, \quad \sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$$

$$T_1 = 423 \text{ K}, \quad T_2 = 300 \text{ K}, \quad D_o = 60 \text{ mm} = 0.06 \text{ m}$$

Outside surface area per 1 meter length of the pipe is

$$A = \pi D_o L = \pi \times 0.06 \times 1 = 0.189 \text{ m}^2$$

The net radiation rate per 1 m length of the pipe is

$$\begin{aligned} Q_r &= e \sigma A (T_1^4 - T_2^4) = 0.8 \times 5.67 \times 10^{-8} \times 0.189 \left(423^4 - 300^4 \right) \\ &= 205 \text{ W/m} \end{aligned} \quad \dots \text{Ans.}$$

Example: Calculate the rate of heat loss from a 6 m long horizontal steam pipe, 60 mm o.d. when carrying steam at 800 kN/m². The temperature of the surrounding atmosphere is 290 K.

Take emissivity, $e = 0.85$ and Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$.

The film coefficient (h_c) for heat loss by natural convection can be calculated by

$$h_c = 1.64 (\Delta T)^{0.25} \text{ W/(m}^2 \cdot \text{K)}$$

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Heat Transfer Operation

Steam is saturated at 800 kN/m² and 443 K (170°C).

Solution : Neglecting the inside resistance and resistance of the metal wall, it may be assumed that the surface temperature of the pipe is 443 K.

Given : $T_1 = 443 \text{ K}$, $T_2 = 290 \text{ K}$, $D_o = 60 \text{ mm} = 0.06 \text{ m}$, $L = 6 \text{ m}$

$$\therefore \Delta T = 443 - 290 = 153 \text{ K}$$

For radiation from the pipe :

$$\begin{aligned}\text{Surface area of the pipe} &= \pi D \cdot L \\ &= \pi \times 0.06 \times 6.0 = 1.131 \text{ m}^2\end{aligned}$$

The rate of heat loss by radiation from the pipe is

$$\begin{aligned}Q_r &= e \sigma A (T_1^4 - T_2^4) \\ &= 0.85 \times 5.67 \times 10^{-8} \times 1.131 (443^4 - 290^4) \\ &= 1714 \text{ W}\end{aligned}$$

The rate of heat loss by convection from the pipe is

$$\begin{aligned}Q_c &= h_c A (T_1 - T_2) \\ &= 1.64 (\Delta T)^{0.25} \times A (T_1 - T_2) \\ &= 1.64 (153)^{0.25} \times 1.131 (443 - 290) \\ &= 998 \text{ W}\end{aligned}$$

$$\begin{aligned}\therefore \text{Total heat loss} &= Q_r + Q_c = 1714 + 998 \\ &= \mathbf{2712 \text{ W}}\end{aligned}$$

Exchange of energy between two parallel plates / planes of different emissivities: (Multiple Reflection Method)

When two non-black bodies are situated at a small distance apart, part of the energy emitted by one body will be reflected back to it by the second body and will then be partly reabsorbed and partly reflected again.

Thus, the heat undergoes a series of internal reflections and absorptions.

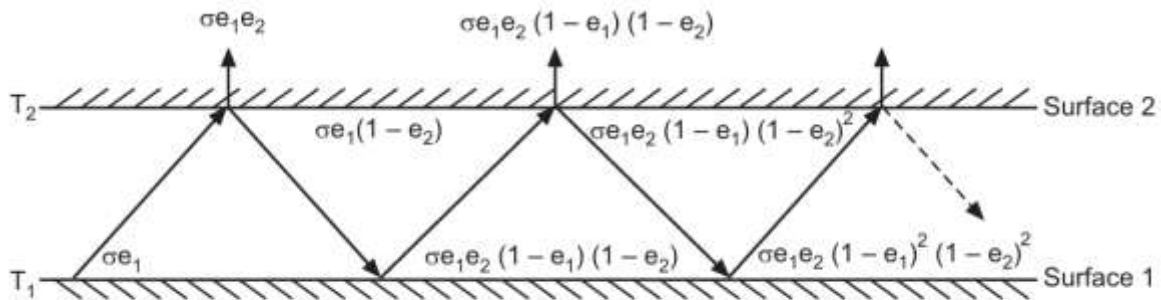
Consider two large gray planes/surfaces that are maintained at absolute temperatures T_1 and T_2 respectively, a small distance apart and exchanging radiation. Let e_1 and e_2 be the emissivities of the surfaces.

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Heat Transfer Operation



**Fig. 4.3 : Radiant heat exchanger between infinite parallel surfaces
(energy originating at surface-1 absorbed by surface-2)**

Consider the energy radiated/emitted from the surface-1

$$\text{energy radiated from surface 1} = \sigma \cdot e_1 T_1^4$$

$$\text{of this, energy absorbed by surface 2} = \sigma e_1 T_1^4 e_2$$

$$\text{and energy reflected by surface 2} = e_1 T_1^4 (1 - e_2)$$

$$\text{of this, energy re-absorbed by surface 1} = \sigma e_1 T_1^4 (1 - e_2) e_1$$

$$\text{and energy re-reflected by surface 1} = \sigma e_1 T_1^4 (1 - e_2) (1 - e_1)$$

$$\text{and of this, energy absorbed by surface 2} = \sigma e_1 T_1^4 (1 - e_2) (1 - e_1) e_2$$

Hence, as a result of each complete cycle of internal reflection, that the absorption is reduced by a **factor $(1 - e_1) (1 - e_2)$** . As the energy suffers an infinite number of reflections, we can write:

Total transfer of energy from surface 1 to surface 2 per unit area and unit time is

$$\begin{aligned} (Q/A)_{1-2} &= \sigma \cdot e_1 e_2 T_1^4 [1 + (1 - e_1) (1 - e_2) + (1 - e_1)^2 (1 - e_2)^2 \dots \text{to } \infty] \\ &= \sigma \cdot e_1 e_2 T_1^4 \frac{1}{1 - (1 - e_1) (1 - e_2)} \\ &= \frac{\sigma \cdot e_1 e_2}{e_1 + e_2 - e_1 e_2} T_1^4 \end{aligned}$$

In a similar manner, considering the radiation emitted by the surface 2, it can be shown that the total transfer of energy from surface 2 to surface 1 per unit area per unit time (i.e., emitted by the surface 2 and absorbed by the surface 1) is

$$(Q/A)_{2-1} = \frac{e_1 e_2 \sigma}{e_1 + e_2 - e_1 e_2} T_2^4$$

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Thus, the net energy transferred per unit area per unit time is

$$\left(\frac{Q}{A}\right)_{12} = \frac{e_1 e_2 \sigma}{e_1 + e_2 - e_1 e_2} (T_1^4 - T_2^4)$$

$$\left(\frac{Q}{A}\right)_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{e_1} + \frac{1}{e_2} - 1}$$

$$\left(\frac{Q}{A}\right)_{12} = \sigma \cdot F_{12} (T_1^4 - T_2^4)$$

$$F_{12} = \frac{1}{\frac{1}{e_1} + \frac{1}{e_2} - 1}$$

(F₁₂ is called overall interchange factor and is a function of e₁ and e₂)

Spheres or cylinders with spherical or cylindrical enclosures:

The net exchange of radiative heat or radiant energy between the inner and outer sphere is given by

$$Q = \frac{\sigma A_1}{\frac{1}{e_1} + \left(\frac{r_1}{r_2}\right)^2 \left(\frac{1}{e_2} - 1\right)} (T_1^4 - T_2^4)$$

$$= \frac{\sigma A_1}{\frac{1}{e_1} + \frac{A_1}{A_2} \left(\frac{1}{e_2} - 1\right)} (T_1^4 - T_2^4)$$

The net exchange of radiative heat or radiant energy between infinitely large concentric cylinders is given by

where A₁ and A₂ are the areas of the inner and outer cylinders/spheres respectively, e₁ and e₂ are the emissivities of the inner and outer cylindrical/spherical surfaces. T₁ and T₂ are the respective temperatures.

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$$Q = \frac{\sigma \cdot A_1 (T_1^4 - T_2^4)}{\frac{1}{e_1} + \frac{A_1}{A_2} \left(\frac{1}{e_2} - 1 \right)} = \frac{\sigma \cdot A_1 (T_1^4 - T_2^4)}{\frac{1}{e_1} + \frac{r_1}{r_2} \left(\frac{1}{e_2} - 1 \right)}$$

$$Q = \sigma A_1 F_{1-2} (T_1^4 - T_2^4)$$

$$F_{12} = \frac{1}{\frac{1}{e_1} + \frac{A_1}{A_2} \left(\frac{1}{e_2} - 1 \right)}$$

- Cengel A. Yunnus. "Heat Transfer – A Practical Approach", McGraw Hill.
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